

# DEXTER FERNANDES

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[github.com/Dexter-Fernandes](https://github.com/Dexter-Fernandes)

## SUMMARY

Computer Vision Engineer with 3+ years building and deploying real-time perception systems in industrial environments. Commercial work spans object detection, classification and tracking, taken from data curation and training in PyTorch through quantisation, edge deployment and on-site validation on NVIDIA and Intel hardware. MSc Robotics (Merit) covering LiDAR SLAM, point cloud processing and sensor fusion in C++17 and ROS2. Works in Python and C++ across multidisciplinary hardware, software and QA teams.

## RELEVANT EXPERIENCE

### Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end machine learning lifecycle for real-time perception on BP drilling rigs, from data curation and training in PyTorch through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Developed and maintained NVIDIA DeepStream 6.x and FFmpeg video pipelines covering camera ingestion, inference and field diagnostics in desert conditions.
- Hardened deployed services with Dockerised microservices, Kafka messaging, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.
- Adapted YOLOv5 detectors in PyTorch, replacing backbone and neck layers and retraining for rig-specific classes, with Optuna sweeps and Weights & Biases tracking across runs.
- Built ingestion, validation and annotation pipelines across image, video and sensor data with Airflow, MLflow, DVC and CVAT, cutting rollout from 72h to under 24h.
- Applied image preprocessing and artefact detection with OpenCV and scikit-image to catch mud and sand contamination on camera feeds, reducing false positives by 13%.

### LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety on live sites in Texas, cutting false positives by 33% within latency and power budgets.
- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes, raising throughput from 5 to 15 FPS under real-time constraints.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising streams and adding ByteTrack multi-object tracking to hold worker and vehicle identities across frames.
- Implemented optical flow based ego-motion estimation feeding CNN classifiers in PyTorch to infer vehicle state, supporting collision avoidance on moving vehicles.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Automated labelling by repurposing existing detection models and added dataset versioning with DVC, cutting manual labelling effort by 80% on large real-world datasets.

## PROJECTS

### DeepStream RTSP Pipeline [github.com/Dexter-Fernandes/deepstream-rtsp-pipeline](https://github.com/Dexter-Fernandes/deepstream-rtsp-pipeline)

06/2026

- Developed an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction, anonymisation and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream.
- Implemented a C++ and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving post-processing into the inference engine.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDF1, ID switches and FPS to guide tracker selection.

### ROS2 SLAM and Nav2 Simulation

02/2026

- Developed and evaluated a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic covering LiDAR-centric SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation.
- Implemented point cloud filtering, segmentation and registration with PCL and Open3D, and added an EKF localisation layer with robot\_localization fusing wheel odometry and IMU.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations to compare mapping and localisation behaviour.
- Developed ROS2 nodes, launch files and parameterised configs in C++ and Python, using tf2 and message synchronisation to debug transform timing issues.

- Benchmarked PyTorch detectors including YOLOv8, YOLOv10, YOLOv11 and RT-DETR on accuracy, latency and model size for edge deployment suitability.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess segmentation and depth estimation under the same edge constraints, testing ONNX and OpenVINO export paths.
- Evaluated Vision Transformer and foundation-model approaches across detection, segmentation, depth estimation and open-vocabulary understanding.

EDUCATION

University of Bristol — MSc Robotics, Merit

09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol.
- Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a PCL point cloud pipeline.
- Ran the full SLAM and perception stack on an NVIDIA Jetson AGX Xavier, working within the platform's compute and memory limits for on-robot operation.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering

07/2016 – 05/2020 · NAGPUR, INDIA

- Foundation in signals, systems, embedded hardware and control, supporting later work in real-time AI, sensor systems and robotics software.

SKILLS

|                        |                                                                                                                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages              | Python, C++17, CUDA C++, Bash, SQL                                                                                                                                                                                                                       |
| Computer vision        | Object detection, image classification, semantic and instance segmentation, object tracking, optical flow, anomaly and defect detection, data augmentation, convolutional neural networks (CNNs), YOLO (v5/v8/v10/v11), RT-DETR, EfficientDet, ByteTrack |
| Deep learning          | PyTorch, TensorFlow, Keras, OpenCV, scikit-image, torchvision, NumPy, Pandas, HuggingFace, scikit-learn                                                                                                                                                  |
| 3D vision and robotics | ROS2 (Robot Operating System 2), rclcpp, rclpy, tf2, LiDAR SLAM, point cloud processing, PCL (Point Cloud Library), Open3D, depth cameras (RealSense D435i), Velodyne VLP-16, sensor fusion, EKF, RTAB-Map, LIO-SAM, Cartographer, Nav2                  |
| Edge and optimisation  | TensorRT (custom plugins, dynamic batching), OpenVINO, ONNX, quantisation-aware training, FP16/INT8, structured pruning, layer fusion, latency and VRAM profiling, NVIDIA Jetson AGX Xavier, Intel Movidius VPU, Intel Core i7 industrial edge           |
| Video pipelines        | NVIDIA DeepStream 6.x/9.0, GStreamer, FFmpeg, RTSP, NvDCF, NvSORT, multi-camera synchronisation, ISP and camera tuning (RGB and IR)                                                                                                                      |
| Data and MLOps         | CVAT annotation, automated labelling, dataset versioning (DVC), Apache Airflow, MLflow, Optuna, Weights & Biases, Apache Kafka, PostgreSQL, benchmarking harnesses, drift tracking, COCO annotation format                                               |
| Production and cloud   | Docker, CI/CD, Git, CMake, pytest, gtest, Microsoft Azure (Container Apps, Blob Storage, queues), REST API design, containerised microservices, autoscaling, monitoring, Linux                                                                           |
| Professional           | Technical leadership, mentoring, Agile and Scrum, Jira, cross-functional delivery                                                                                                                                                                        |

ADDITIONAL EXPERIENCE

GrowthStage — Technical Assessor, Computer Vision and Software Engineering

05/2026 – PRESENT · REMOTE

Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019)

Best Speaker award, ABC Toastmasters Club (01/2020)